Ishaq Hamza

linkedin.com/in/ishaqhamza ihamza@purdue.edu

OVERVIEW

Operations Research PhD student at **Purdue IE**, **Math & Computing undergrad** at **Indian Institute of Science**. Working on **reinforcement learning** theory. Aiming to design nuanced, robust and sample-efficient learning and control algorithms for challenging problems in high-noise settings.

EDUCATION

Purdue University: PhD in Operations Research 2026–30

Advisor: Dr. Zaiwei Chen

Other PhD Offers: Georgia Tech ISyE, **Columbia University** IEOR.

Awardee of the prestigious *Dr. W. Dale and Mrs. Jeanne C. Compton Award* for four years of doctoral study, over and above standard PhD support.

Research Focus: Reinforcement Learning, Stochastic Games, Optimization Theory, Stochastic Modeling.

Indian Institute of Science (IISc): B.Tech in Math and Computing 2022–26

CGPA: **3.96/4**, **Rank 4**

Relevant Coursework:
Online and Reinforcement Learning || **Game Theory** || **Measure-Theoretic Probability** || **Regression and Time-Series Analysis** || **Stochastic Finance** || **Applied DS&AI** || **Renewal and Queueing Theory** || **Computational Optimization** || **Detection & Estimation Theory** || **Numerical Methods** || **Nonlinear Systems & Stability** || **Advanced Real Analysis**

EXPERIENCE (Undergrad)

Summer Research Intern May–August 2025

Dr. Zaiwei Chen, Purdue IE *Optimal Single-Timescale Actor-Critic Algorithm*

- Proposed two variants of the **actor-critic algorithm** with **single-timescale updates** and achieved the **optimal sample complexity** in the **off-policy** setting through a tight analysis under **provably minimal assumptions**.
- Used advanced mathematical tools such as **Moreau envelopes**, **stochastic approximation**, and **Markov chain mixing** to achieve the results. **(Published at the NeurIPS 2026 MLxOR workshop.)**

Teaching Assistant (Undergrad Junior Year) Jan–Apr 2025

Advanced Convex Optimization *Dr. Kunal N. Chaudhury, IISc EE* | course webpage

- Conducted **tutorials**, answered questions, and graded **assignments** for a **class of PhD students**.
- Covered rigorous **convex analysis** in extended-real settings and the **operator-theoretic** treatment of optimization algorithms in both smooth and **non-smooth** settings, using set-valued analysis for **proximal algorithms**.

SKILLS

Theory: Stochastic Approximation, Stochastic Modeling, Statistical Inference, Game Theory, Linear Models, Decision Trees and Neural Networks.

LaTeX, R, Python (NumPy, Pandas, PyTorch).

PUBLICATIONS

- Achieving ϵ^{-2} Sample Complexity in Single-Loop Actor-Critic, NeurIPS MLxOR 2026, **Ishaq Hamza**, Zaiwei Chen. **Undergrad Publication**, arXiv link

UNDERGRADUATE COURSE PROJECTS

Extending Soft Actor-Critic to Multi-Agent RL — **Spring 2025**

RL Course Project — GitHub Link

- Surveyed **Entropy-Regularized RL** and **Soft Actor-Critic algorithms** in several settings.
- Proposed and implemented a **novel multi-agent extension** of the algorithm, incorporating **Cross-Entropy Regularization**.
- Extended the analysis and proved **single-step evaluation** and **improvement theorems**. Provided a **counterexample** to global optimal convergence.
- Proposed **Dynamic Cross-Entropy Regularization** to escape **sub-optimal Nash equilibria** and designed experiments to validate it (successfully implemented, with positive results, by the rest of the team).

Game-Theoretic Market-Making and Trading — **Spring 2025**

Game Theory Course Project

- Surveyed **10+ papers** on **modeling and equilibrium analysis** of **market makers and traders** in financial markets.
- Surveyed **5+ papers** on **information asymmetry** and its impact on market making.
- Formulated a **novel generalized model** of trading with **priced information exchange**.

Automated Equity Research Analyst — **Fall 2024**

Applied DS&AI Course Project **(Team of 6)**, Contributions: — GitHub Link

- Conceived the **rating-generation mechanism** and project structure for a **team of 6**.
- Implemented a **self-engineered variant of the multiplicative weights algorithm** to generate ratings for stocks from **aspect-based sentiment analysis scores** computed on financial news data.
- Tested on both **real and geometric-Brownian-motion-simulated prices**, with **high-noise expert ratings**.

Financial Network Modeling — **Fall 2024**

Data Analytics Course Project **(Led a team of 5)** — GitHub Link

- Researched **generative models** to study and analyze **graph community structures**.
- Studied relationships in stock price movements using a **hierarchical weighted stochastic block model**.
- Applied the same model to **import–export data** across 200+ countries to study **trade relations**, obtaining the expected results.

AWARDS & ACHIEVEMENTS

- **Dr. W. Dale and Mrs. Jeanne C. Compton Award**: four-year fellowship recipient.
- **Simon Marais Mathematics Competition (SMMC) 2024**: **Global 98th** (Institute 5th) ‖ results
- **Indian Statistical Institute Entrance Test 2022**: National **Rank 62**.
- **Dr. Kolar & Mrs. Girija Kodandapani Scholarship**, awarded to **one student per batch**. ‖ odaa.iisc.ac.in
- **ITTIAM Fellowship**: two-time awardee (2024–25 and 2025–26). ‖ odaa.iisc.ac.in